A3-PS015-TC142

Explainable Home & Last-Mile Connectivity Diagnosis

An AI-powered system that identifies likely connectivity causes and gives users clear, actionable next steps.

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AIORI Hackathon 3 • October 14, 2026

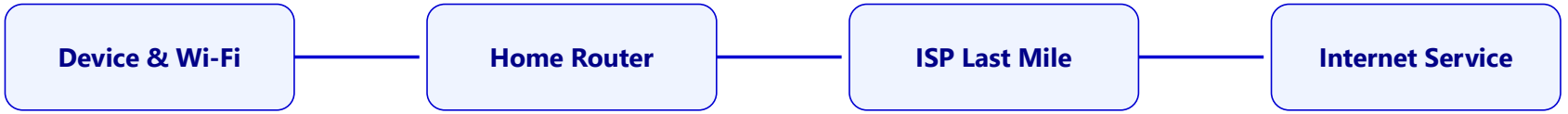
Problem Description

Users can see that their connection is poor — but rarely why.

- Slow speeds, buffering, high latency, packet loss, and dropouts disrupt work, learning, calls, and gaming.
- The root cause can be Wi-Fi signal weakness, interference, congestion, router faults, or an ISP/last-mile issue.
- Conventional speed tests report metrics but do not explain the probable cause or what the user should do next.
- This leads to blind restarts, unnecessary setting changes, and avoidable ISP support calls.

Background

Home connectivity is a multi-layer experience.



- The same symptom can originate at different layers, so a single speed value is not enough for diagnosis.
- Useful observations include latency, jitter, packet loss, download/upload throughput, Wi-Fi signal, and time patterns.
- A helpful tool must translate measurements into confidence-aware, human-readable guidance.

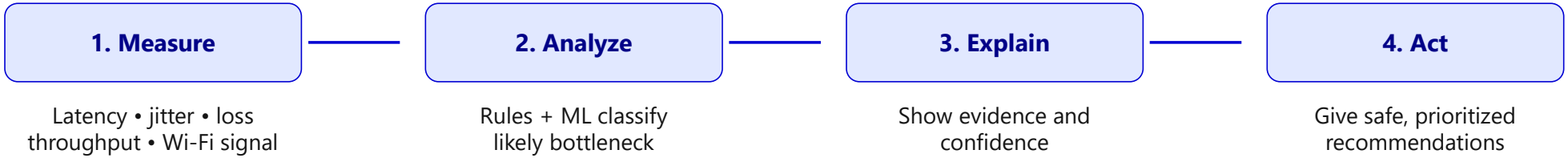
Literature & Standards Review

What existing measurement practice gives us — and what it misses.

Source / Practice	Contribution	Gap / Opportunity
Speed tests	Download/upload throughput; ping	Usually no root-cause explanation
RFC 6349	TCP throughput testing framework	Metrics need interpretation for home users
RFC 2681	One-way delay measurement concepts	Delay alone does not isolate the failing layer
Our approach	**Multi-metric diagnosis + explanations**	**Actionable, personalized troubleshooting**

Proposed Solution

An explainable diagnosis engine turns raw network tests into practical advice.



- Example: weak signal plus normal wired performance → “Move closer to the router or use 5 GHz.”
- Example: healthy local link but persistent external loss/latency → “Collect evidence and investigate ISP-side issues.”

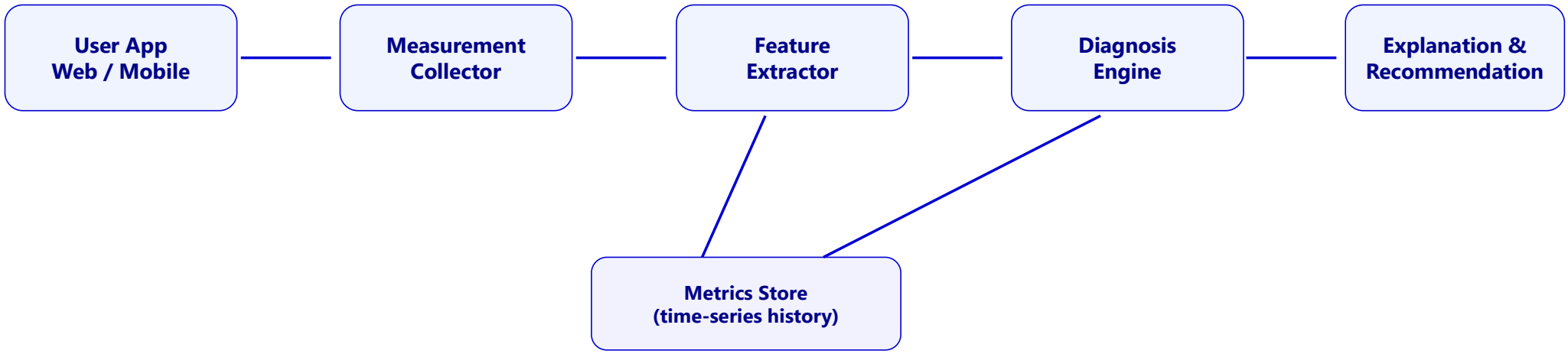
Optimization Proposed

From raw metrics to a diagnosis users can trust.

Before	Enhancement	After	User value
Single snapshot	Repeated tests over time	Separates transient congestion from persistent faults	
One aggregate score	Layer-aware feature set	Distinguishes Wi-Fi, router, and ISP signals	
Opaque classification	Evidence + confidence display	Makes the diagnosis explainable	
Generic advice	**Ranked, safe recommendations**	**Reduces unnecessary resets and support calls**	

Solution Architecture & Design

Component view

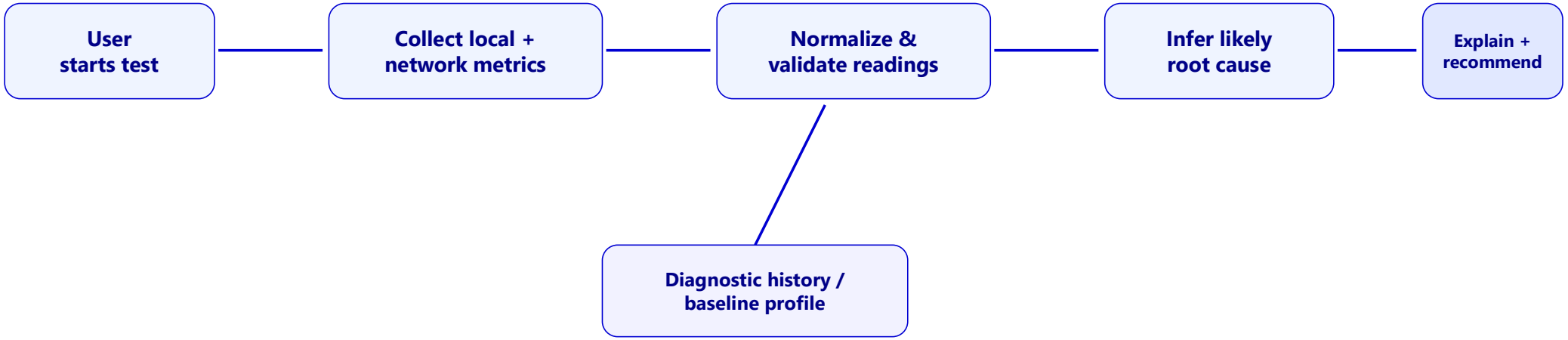


Inputs: latency, jitter, packet loss, download/upload, signal strength, test history

Figure: Explainable Connectivity Diagnosis Architecture

Data Flow Diagram

Data flow from a connectivity test to a user decision.

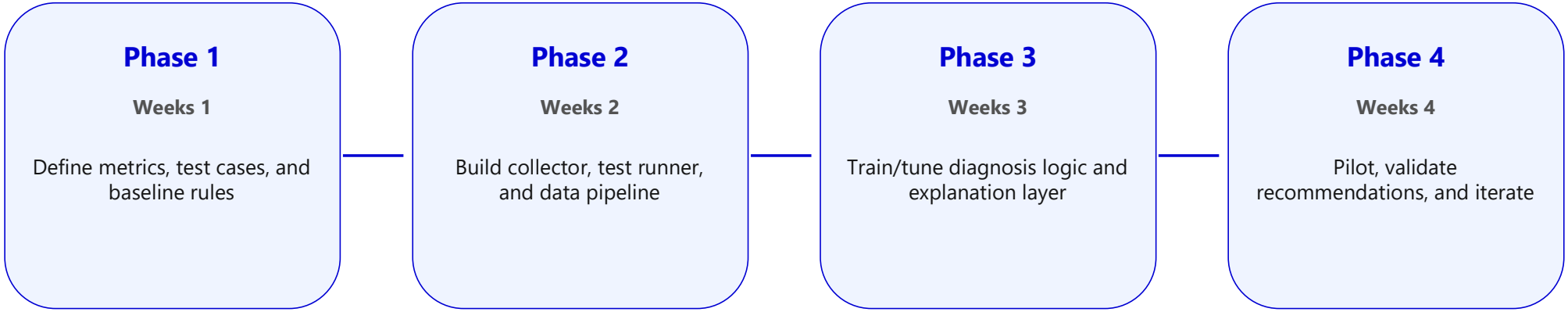


Output includes: likely cause • supporting evidence • confidence • next-best action

Figure: Data Flow Diagram (DFD)

Deployment Timeline

Phased delivery keeps the system measurable, safe, and useful.



Success measures: diagnosis accuracy • explanation usefulness • recommendation completion • reduced unnecessary support contacts

References

Selected references supporting network measurement concepts.

[1] IETF RFC 6349. Framework for TCP Throughput Testing. August 2011.

[2] IETF RFC 2681. A Round-trip Delay Metric for IPPM. September 1999.

[3] IETF RFC 3393. IP Packet Delay Variation Metric for IPPM. November 2002.

[4] ITU-T Recommendation Y.1540. Internet protocol data communication service — IP packet transfer and availability performance parameters.

Thank You

Making home connectivity problems understandable,
explainable, and easier to resolve.

Questions?